

Dependency Architecture and Critical Path

Contract interfaces, dependencies, parallelization and architectural decisions

ENTERPRISE REMEDIATION REPORT

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Evidence-led | contract-governed | accessibility-aware | rollout-ready

Truthful workflows. Explicit controls. Verifiable enterprise evidence.

Status: planned **Rule:** a downstream module may adopt a contract only after its owner, interface, tests, telemetry, rollout, and rollback are approved.

Program dependency graph

Architecture diagram source

```

flowchart LR
  G1["G0-01 Current-truth baseline"] --> G2["G0-02 ADR and ownership freeze"]
  G1 --> G3["G0-03 Role fixtures and certification baseline"]
  G2 --> A3["A-03 DisplayContext"]
  G2 --> A2["A-02 Completeness and error taxonomy"]
  G2 --> A5["A-05 Action/export contract"]
  G2 --> A6["A-06 Public/auth WCAG"]
  G2 --> A8["A-08 Capability and entitlement parity"]
  A3 --> A4["A-04 Currency and locale migration"]
  A3 --> A1["A-01 Inventory headline truth"]
  A2 --> A1
  A5 --> A1
  A6 --> A7["A-07 Accessible auth and funnel"]
  A8 --> A9["A-09 Safe immediate aliases"]
  A8 --> B1["B-01 Canonical route registry"]
  B1 --> B2["B-02 Capability-derived discovery"]
  A2 --> B3["B-03 Page-state registry"]
  G2 --> B4["B-04 Token and appearance contract"]
  B4 --> B5["B-05 Top-20 theme migration"]
  G3 --> B6["B-06 Certification manifest/harness"]
  B1 --> B6
  A7 --> B7["B-07 Resumable onboarding"]
  B3 --> C0["C-00 Shared data/form/dialog/copy primitives"]
  A5 --> C0
  B4 --> C0
  B6 --> C0
  A1 --> C1["C-01 Inventory wave"]
  C0 --> C1
  C1 --> C2["C-02 Purchasing/AP wave"]
  C2 --> C3["C-03 Finance/reconciliation/close wave"]
  C3 --> C4["C-04 HRIS/payroll wave"]
  C4 --> C5["C-05 Settings wave"]
  C5 --> C6["C-06 POS wave"]
  C0 --> C7["C-07 Characterization-led decomposition"]
  C7 --> C1
  C7 --> C2
  C7 --> C3
  C7 --> C4
  C7 --> C5
  C7 --> C6
  C6 --> D1["D-01 Automated certification"]
  D1 --> D2["D-02 Human/domain validation"]
  D1 --> D3["D-03 Performance and observability"]
  D2 --> D4["D-04 Cohort rollout and support"]
  D3 --> D4
  D4 --> D5["D-05 Final assurance decision"]

```

Critical path interpretation

The longest control-dependent path is:

G0-01 → G0-02 → A-08 → B-01 → B-02/B-06 → C-00 → C-01 → C-02 → C-03 → C-04 → C-05 → C-06 → D-01 → D-02/D-03 → D-04 → D-05.

Why this path dominates:

- 1 capability and route truth determine which users and packages may reach every later module surface;
- 1 the certification manifest cannot be complete until canonical routes and roles are known;
- 1 module waves must adopt stable shared contracts rather than inventing local variants;
- 1 final certification must run against the migrated canonical surface, not aliases or temporary UI;
- 1 rollout cannot begin until automated and human validation evidence converge.

The financial/display-truth path runs in parallel but is also Phase-A blocking:

G0-02 → A-03 → A-04 and G0-02 → A-02 → A-01.

Neither path can be waived. Phase A closes only when both are complete.

Safe parallel work

After G0-02 freezes interfaces and owners:

- 1 A-03 DisplayContext, A-05 action/export, A-06 WCAG, and A-08 capability design may run concurrently.
- 1 A-01 may begin service characterization while A-03/A-02 are finalized, but UI cutover waits for both contracts.
- 1 B-03 page states and B-04 semantic tokens may run concurrently.
- 1 B-06 may scaffold fixtures and manifest schema before B-01, but route population waits for the registry.
- 1 Module teams may create characterization tests during Phase B; production migrations wait for C-00.
- 1 D-03 performance instrumentation can be designed early; production thresholds are accepted only after representative route data exists.

Unsafe parallelization:

- 1 do not migrate money displays before DisplayContext and approved currency semantics;
- 1 do not redirect duplicate routes before canonical route and analytics decisions;
- 1 do not hard-enforce entitlements globally before legacy tenant and subscription migrations;
- 1 do not split high-risk components before characterization tests;
- 1 do not create per-module table, state, form, or theme primitives while the shared contract is unsettled;
- 1 do not certify screenshots produced before stable fixtures, route ownership, and supported theme decisions.

Target contract register

CT-01 — DisplayContext

- 1 **Purpose:** provide trusted organization locale, currency, timezone, number/date formatting policy, organization ID, and optional location scope.
- 1 **Owner:** platform/domain architecture; finance control approves money semantics.
- 1 **Boundary:** server-derived and passed through read models; client consumes immutable display metadata.
- 1 **Current foundation:** organization currency/timezone fields; dashboard, inventory history, AP history, reports, and daily-habit services already use parts of this context. `lib/i18n/formatters.ts` still defaults to USD.
- 1 **Dependencies:** G0-02 decisions on organization truth; no schema change unless existing fields prove insufficient.
- 1 **Consumers:** inventory, purchasing/AP, finance, POS, notifications, reports, analytics, settings, exports.
- 1 **Version/migration:** introduce `v1`, adapter legacy formatters, migrate risk-tier domains, then ban unapproved literals.
- 1 **Failure:** fail closed for financial displays or render explicit unavailable/partial state; never substitute USD silently.
- 1 **Telemetry:** missing/invalid context count, fallback use count, display-context version.
- 1 **Tests:** locale/currency fixtures, invalid timezone, organization-scope, SSR/client parity, notification/export parity.
- 1 **Rollout/rollback:** feature flag by module; rollback to adapter, never to silent default.
- 1 **Adoption measure:** certified business UI has zero unapproved currency fallback.

CT-02 — Service-owned UI read model

- 1 **Purpose:** make aggregation, scope, period/as-of, completeness, provenance, and policies server-owned.
- 1 **Owner:** domain backend/data owner; finance/control validates metric definitions.
- 1 **Boundary:** service returns typed read model; UI performs no organization-wide aggregation over paginated rows.
- 1 **Current foundation:** dashboard read model, inventory movement history, inventory stats, AP history, finance and assurance models.

- 1 **Dependencies:** CT-01 and CT-05 completeness/error taxonomy.
- 1 **Consumers:** every KPI, table summary, proof banner, export, and command-center surface.
- 1 **Version/migration:** add fields compatibly, build adapters, compare old/new in shadow mode, cut over route-by-route.
- 1 **Failure:** explicit error/partial/stale/unavailable result with correlation ID.
- 1 **Telemetry:** generation latency, source completeness, snapshot age, comparison drift.
- 1 **Tests:** page-size invariance, filter equality, tenant isolation, monetary precision, failure paths, as-of behavior.
- 1 **Rollout/rollback:** shadow results before visible cutover; retain previous read path behind short-lived rollback flag.
- 1 **Adoption measure:** 100% headline metrics disclose scope/currency/as-of/completeness.

CT-03 — Capability decision

- 1 **Purpose:** combine permission, module entitlement, organization lifecycle, location scope, access intent, and dependency state.
- 1 **Owner:** security/IAM and SaaS platform/billing.
- 1 **Boundary:** server-authoritative decision; UI consumes presentation-safe state/reason.
- 1 **Current foundation:** RBAC, module catalog/evaluator, explicit mode: "enforce" paths, API module access, audit logging.
- 1 **Gap:** default module mode is observe; hard-enforcement flag is false; 257 candidates remain.
- 1 **Dependencies:** G0-02 entitlement-source ADR and legacy tenant migration plan.
- 1 **Consumers:** route guards, actions, APIs, jobs, navigation, search, command palette, shortcuts, notifications, exports.
- 1 **Version/migration:** observe → compare → pilot enforce → cohort enforce → default enforce; never big-bang.
- 1 **Failure:** denied/locked/read-only/not-configured with stable reason code; no data leak in explanation.
- 1 **Telemetry:** would-block vs deny, decision mismatch, bypass attempt, dependency gap, legacy-default usage.
- 1 **Tests:** role/package/location matrix, wildcard behavior, read-only intent, deep links, server/UI parity, tenant boundaries.
- 1 **Rollback:** return selected cohorts to observe while retaining audit logging and server RBAC.
- 1 **Adoption measure:** all registered protected surfaces have explicit tested decisions.

CT-04 — Canonical route registry

- 1 **Purpose:** one route per user intent with aliases, owner, module, capability, breadcrumb, telemetry, state contract, and retirement date.
- 1 **Owner:** enterprise UX/frontend platform.
- 1 **Boundary:** typed config consumed by server routes and client navigation; server redirects preserve locale/query.
- 1 **Current foundation:** 81-link sidebar, localization helpers, permission-aware shell, some safe redirects.
- 1 **Dependencies:** CT-03 capability decision.
- 1 **Consumers:** routes, sidebar, navbar search, shortcuts, command palette, sitemap/support docs, analytics, certification manifest.
- 1 **Version/migration:** register current routes; choose canonical; mark aliases; instrument traffic; redirect; retire implementation after evidence window.
- 1 **Failure:** unresolved route is excluded from release certification and logged as registry debt.

- 1 **Telemetry:** alias traffic, redirect errors, destination denials, broken bookmarks.
- 1 **Tests:** uniqueness, capability presence, locale/query preservation, analytics continuity, no redirect loops.
- 1 **Rollback:** restore alias implementation/redirect mapping without losing canonical telemetry.
- 1 **Adoption measure:** 100% active routes registered; no duplicate active task implementation.

CT-05 — Page-state union

- 1 **Purpose:** distinguish loading, empty, filtered empty, partial, stale, error, permission denied, locked, not configured, and success.
- 1 **Owner:** design system and backend error-contract owners.
- 1 **Boundary:** services emit stable state/reason metadata; shared UI renders accessible recovery behavior.
- 1 **Current foundation:** `DashboardRouteState` already covers nine important states; BI and transaction-history components include partial/stale semantics.
- 1 **Gap:** no complete registry/adoption, limited localization, no filtered-empty/not-configured distinction everywhere, many manual messages.
- 1 **Dependencies:** error taxonomy and CT-03 capability reason codes.
- 1 **Consumers:** every canonical page and module workbench.
- 1 **Version/migration:** extend current primitive, localize, register route requirements, migrate risk tier first.
- 1 **Failure:** unknown states map to safe generic error with correlation, never successful empty.
- 1 **Telemetry:** state distribution, retries, recovery success, repeated failures.
- 1 **Tests:** component semantics, focus/announcement, recovery actions, service-state mapping.
- 1 **Rollback:** route-level adapter to prior explicit state; no silent catch-to-empty.
- 1 **Adoption measure:** every canonical route declares and tests its state contract.

CT-06 — Action capability and artifact contract

- 1 **Purpose:** ensure every visible action is hidden, locked, disabled with reason, or executable with a verifiable result.
- 1 **Owner:** backend/integration plus audit/evidence owner.
- 1 **Boundary:** server declares permission/entitlement/idempotency/result type; client renders state and tracks outcome.
- 1 **Current foundation:** working PO and reconciliation exports, action wrappers, report trust/export gate.
- 1 **Gap:** dead inventory/user export controls and simulated customer export success.
- 1 **Dependencies:** CT-03 and evidence artifact semantics.
- 1 **Consumers:** buttons, menus, bulk actions, exports, background jobs, AI proposals.
- 1 **Version/migration:** inventory actions; disable false controls; standardize artifact response; migrate exports/actions.
- 1 **Failure:** no success toast without result; failures return correlation and recovery path.
- 1 **Telemetry:** invocation/result/artifact IDs, row counts, latency, failure/retry, idempotency collisions.
- 1 **Tests:** artifact exists, metadata/filter parity, denied/disabled behavior, retries, duplicate invocation.
- 1 **Rollback:** disable control or return to known working exporter; never simulate completion.
- 1 **Adoption measure:** zero enabled controls without real outcomes.

CT-07 — Accessible form field

- 1 **Purpose:** standardized label, hint, error, invalid state, autocomplete, focus, live announcement, and localization behavior.
- 1 **Owner:** design system/accessibility; auth and domain teams consume.
- 1 **Boundary:** reusable field primitives and server-error adapter; no sensitive error leakage.
- 1 **Current foundation:** form libraries and visible validation exist.
- 1 **Gap:** active auth forms have no programmatic validation relationships or autocomplete.
- 1 **Dependencies:** content/legal decision for auth fields and CT-05 errors.
- 1 **Consumers:** auth, settings, items, purchasing, accounting, HRIS/payroll.
- 1 **Version/migration:** auth first; shared primitive; high-risk forms; remainder by module wave.
- 1 **Failure:** focus first invalid control, announce summary, preserve entered safe data.
- 1 **Telemetry:** validation category, submit failure, completion time without sensitive values.
- 1 **Tests:** axe, keyboard, screen reader, browser autofill, server/client error mapping, EN/FR.
- 1 **Rollback:** retain prior form behind route flag while fixing blocking accessibility defects.
- 1 **Adoption measure:** all certified forms meet WCAG 2.2 AA manual and automated gates.

CT-08 — Enterprise data view

- 1 **Purpose:** shared filters, sorting, paging, selection, bulk safety, responsive policy, saved views, export parity, and evidence metadata.
- 1 **Owner:** design system/data UX.
- 1 **Boundary:** server query/read model owns data and totals; component owns interaction/accessibility.
- 1 **Current foundation:** DataTable, transaction-history shell, mobile card pattern, command primitives.
- 1 **Dependencies:** CT-02, CT-05, CT-06, CT-07, CT-09.
- 1 **Consumers:** inventory, AP, finance, reconciliation, accounting, HRIS/payroll, settings, POS history.
- 1 **Version/migration:** define column priority and adapter API; pilot inventory; migrate module waves.
- 1 **Failure:** preserve filters/selection safely; bulk actions fail atomically or report per-row result explicitly.
- 1 **Telemetry:** query latency, filter use, overflow/mobile fallback, export parity, bulk failures.
- 1 **Tests:** keyboard table semantics, mobile identity/actions, pagination totals, saved views, export filters.
- 1 **Rollback:** adapter retains legacy table behind feature flag without altering service truth.
- 1 **Adoption measure:** dense certified routes share the contract and preserve supported mobile tasks.

CT-09 — Semantic design-token and appearance contract

- 1 **Purpose:** one semantic color/density/spacing/typography API across light, dark, and system behavior.
- 1 **Owner:** design-system/brand.
- 1 **Boundary:** global tokens define semantics; components consume them; exceptions are registered.
- 1 **Current foundation:** dashboard variables and a strong dark identity.
- 1 **Gap:** 91 forced-dark lines, 72 manual-color files, alternate theme islands, and an appearance control that can disagree with rendering.
- 1 **Dependencies:** product decision on supported appearance modes.
- 1 **Consumers:** all application and public UI.

- 1 **Version/migration:** freeze token names; add CI exception register; migrate top 20 routes; then module waves.
- 1 **Failure:** unsupported mode is not offered; invalid token falls back to accessible semantic default.
- 1 **Telemetry:** user mode, route mismatch, contrast regression, exception count.
- 1 **Tests:** visual/contrast snapshots per supported theme, high contrast, reduced motion, zoom.
- 1 **Rollback:** module-level theme adapter; never strand mixed unreadable modes.
- 1 **Adoption measure:** preference matches rendering; unregistered literal colors trend to zero.

CT-10 — Certification manifest

- 1 **Purpose:** define and record route, role, locale, browser, viewport, package, fixture, theme, accessibility, performance, screenshot, and reviewer evidence.
- 1 **Owner:** QA/release assurance.
- 1 **Boundary:** machine-readable manifest drives tests and stores results; humans sign named validation fields.
- 1 **Current foundation:** Playwright specs, browser smoke scripts, screenshots, axe use, release gates.
- 1 **Gap:** breadth, reusable role fixtures, three-browser coverage, and expert sign-off are incomplete.
- 1 **Dependencies:** CT-04 route registry, CT-03 roles/capabilities, G0-03 fixtures.
- 1 **Consumers:** CI, release gates, QA, accessibility, finance/control, support.
- 1 **Version/migration:** schema first; public and top-20 routes; all canonical routes; historical evidence retention.
- 1 **Failure:** missing blocking evidence prevents certification; advisory gaps require accepted residual risk.
- 1 **Telemetry:** pass/fail/skip age, flaky rate, route coverage, reviewer status, evidence expiry.
- 1 **Tests:** schema validation, manifest completeness, fixture redaction, artifact existence, result freshness.
- 1 **Rollback:** pause promotion; retain last certified version and evidence; never fabricate pass status.
- 1 **Adoption measure:** every certified release has complete, fresh, reviewer-backed evidence.

Architecture decisions required at Gate 0

ADR-UI-001	DisplayContext source, validation, fallback, and versioning	A-03, A-04, A-01
ADR-UI-002	Inventory value/profit metric definitions	A-01
ADR-UI-003	Entitlement source and observe-to-enforce policy	A-08, B-01, B-02
ADR-UI-004	Canonical route selection and deprecation window	A-09, B-01
ADR-UI-005	Error/completeness taxonomy and correlation policy	A-02, B-03
ADR-UI-006	Action artifact schema and evidence retention	A-05, C-00
ADR-UI-007	Supported appearance modes and token exception policy	B-04, B-05
ADR-UI-008	Minimum registration fields and onboarding lifecycle	A-07, B-07
ADR-UI-009	Supported browser/viewport/screen-reader matrix	B-06, D-01, D-02
ADR-UI-010	Release authority and residual-risk acceptance	D-04, D-05